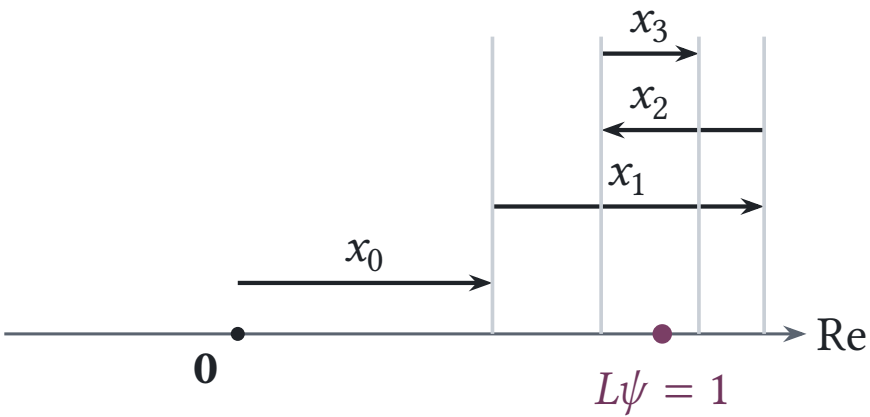
(a) scalar reflection: $X = \mathbb{C}$, $\psi = 1$

one row per exit

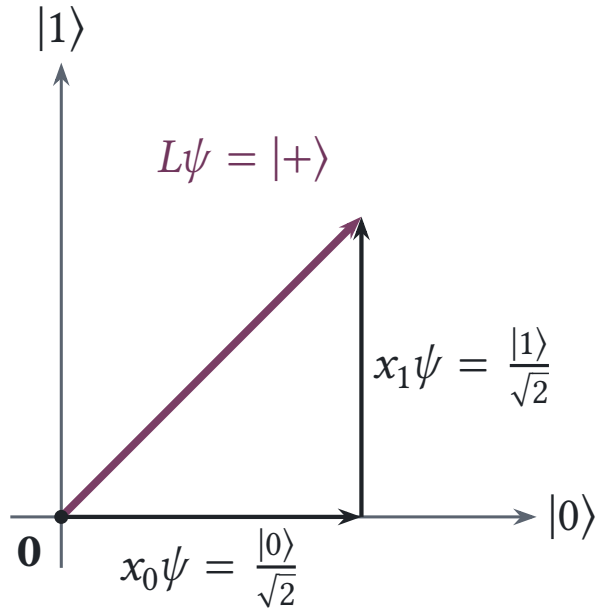


$x_0 = \frac{3}{5}, \quad x_1 = \frac{16}{25}, \quad x_2 = -\frac{48}{125}, \quad x_3 = \frac{144}{625}$

The partial sums converge to 1.

$$\rho_{\text{meas}} = \rho_{\text{coh}} = 1$$

(b) three-cycle: $X = \mathbb{C}^2$, $\psi = |+\rangle$



exits laid head to tail;
real amplitudes in the basis $|0\rangle, |1\rangle$

$$\rho_{\text{meas}} = I/2 \quad \neq \quad \rho_{\text{coh}} = |+\rangle\langle+|$$